

Duy Linh Hoang

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Portfolio: bigbodydubot.github.io/WebPortfolio

Professional Profile

Computer Science student with a Chemistry minor, graduating December 2026. Develops software, analyzes data, investigates security controls, and documents laboratory experiments through academic projects. Brings customer service and leadership experience in staff training, sales analysis, scheduling, and cash reconciliation.

Education

Montclair State University | Montclair, NJ

B.S. in Computer Science | Chemistry Minor | Expected Graduation: December 2026

Dean's List: Fall 2024 through Spring 2026

Middlesex Community College | Edison, NJ

Associate of Liberal Arts | Member of the 2021–2022 wrestling program

Skills

Programming and systems: Python, C, SQL, SQLite, NumPy, Matplotlib, Linux, POSIX processes, threads, semaphores, sockets, and message queues.

Data and laboratory: Excel, StatCrunch, descriptive statistics, regression, calibration, measurement uncertainty, UV-Vis spectrophotometry, acid-base titration, and pH measurement.

Networks and security: Wireshark, Cisco Packet Tracer, Docker lab environments, Scapy, iptables, Linux permissions and ACLs, and SQL injection and CSRF analysis.

Business and communication: Staff training, customer service, sales reporting, labor scheduling, cash reconciliation, technical writing, presentations, and Toast POS.

Work History

The Chop Shop | Old Bridge, NJ | Server | July 2026–Present

- Explain menu options, preparation choices, and beverage pairings so guests can make informed selections; provide attentive service throughout the dining experience.

Meximodo | Jersey City and Metuchen, NJ | Head Server | October 2023–May 2026

- Trained servers on service steps, allergy protocols, menu knowledge, and sales techniques to support consistent service; gave management feedback on new hires' readiness.
- Used Toast POS for order entry, payments, and menu updates, supporting accurate transactions and coordination of service tasks.

Madison Modern Social | Old Bridge, NJ | Server | February 2025–August 2025

- Supported the grand opening by training staff, assisting with menu rollouts, and serving launch events; helped the team prepare for high-volume service and changing priorities.

Red Robin | Woodbridge, NJ | Associate Manager and Server | November 2020–April 2023

- Reviewed sales trends and labor costs to inform staffing schedules and operating decisions; reconciled cash and deposits and flagged discrepancies for management review.
- Tracked comps, voids, discounts, and refunds to identify transaction patterns and support oversight of revenue and payment controls.

Duy Linh Hoang | Selected Academic Projects

Reports, code, calculations, and lab setup downloads:

bigbodydubot.github.io/WebPortfolio

Software and Databases

Student Enrollment Application | Python and SQLite

- Built a command-line application to search courses, enroll or withdraw students, and manage records. Used parameterized SQL and a composite enrollment key to organize database operations and represent course membership consistently.
- Produced an application, database, and schema documentation connecting database constraints and application logic to student record workflows.

HTTP Server and Client | Python and TCP Sockets

- Implemented a web server that parses GET requests, serves files, and returns HTTP 200 or 404 responses. Built a client and test page demonstrating socket-based request-and-response communication.

Data Analysis

Statistical Summaries and Regression | StatCrunch

- Created histograms, box plots, regression models, and residual analyses to examine distributions and relationships in supplied datasets.
- Documented results and model limitations, including weak predictive relationships in the analyzed data; distinguished observed associations from dependable predictions.

Cybersecurity and Systems

Linux Access Controls and Network Security | Linux, Docker, Scapy, and iptables

- Tested file ownership, permissions, and ACLs; documented how least-privilege decisions affect resource access.
- Ran packet sniffing, spoofing, and firewall experiments in controlled lab environments; compared filtering and connection-tracking behavior and produced reports explaining the observed controls and their limitations.

Concurrency and Interprocess Communication | C and POSIX

- Implemented message queues, signal-based timers, and thread synchronization using semaphores. Produced code and outputs demonstrating process communication and coordinated access to shared resources.

Chemistry and Laboratory Analysis

Spectrophotometric Calibration | UV-Vis and Excel

- Prepared ferricyanide standards, measured absorbance at 420 nm, and compared external calibration with standard addition to translate instrument readings into concentration estimates.
- Produced a report and calculation workbook documenting differing estimates, uncertainty, and possible matrix effects; evaluated the measurement methods without assuming calibration fit alone established accuracy.

Titrant Standardization and Unknown Acid Analysis | Titration and pH Measurement

- Standardized sodium hydroxide against potassium hydrogen phthalate and measured an unknown acid's pH during titration; established the titrant concentration and collected detailed readings near rapid pH changes.
- Created a titration curve and laboratory report linking solution preparation, measurements, and endpoint interpretation; documented the experimental basis for evaluating the sample.